Goddard 26–27 Flight Model

Governing Equations

ITE Goddard Team

Version test

This document states every equation implemented in the `goddard` Python package, so the physics can be reviewed without reading source code. Each equation carries the module that implements it and a confidence tag:

- [E] Established result. Analytic, or a long-standing published correlation, implemented directly.
- [A] Engineering approximation. Right form and magnitude; the leading constants are calibrated, not derived.
- [U] **Not validated.** Used because something is needed, but not yet checked against data or a trusted tool.

Anything tagged **[U]** must not be relied on for an absolute number without the cross-check named in Section 11.

Contents

1 Notation **2**

2 Atmosphere **2**

3 Propellants **3**
3.1 Nitrous oxide, saturated . . . 3
3.2 Fuel blend . . . 3

4 Motor **4**
4.1 Self-pressurizing tank blowdown . . . 4
4.2 Showerhead injector . . . 4
4.3 Grain and burnthrough . . . 5
4.4 Nozzle . . . 5
4.5 Chamber pressure balance . . . 6

5 Aerodynamics **6**
5.1 Geometry . . . 6
5.2 Drag build-up . . . 6
5.3 Normal force and centre of pressure . . . 7
5.4 Roll from the 1° fin cant . . . 7

6 Structures **8**
6.1 Sandwich fin stiffness . . . 8
6.2 Flutter and divergence . . . 8
6.3 Nose-tip heating . . . 9

7 Recovery **9**

8 Mass properties **10**

9 Flight dynamics **10**

10 Calibration constants and band mode **11**

11 Validation status **11**

12 References **12**

1 Notation

Symbol	Meaning	Symbol	Meaning
ρ	density	P	pressure
T	temperature	a	speed of sound
μ	dynamic viscosity	M	Mach number
V	speed	q	dynamic pressure
α	angle of attack	θ	body pitch from vertical
γ	ratio of specific heats	Γ	flight path angle
p	roll rate	δ	fin cant angle
$\dot{m}$	mass flow	$\dot{r}$	regression rate
G_{ox}	oxidiser mass flux	O/F	oxidiser/fuel ratio
c^*	characteristic velocity	C_F	thrust coefficient
A_t	throat area	ϵ	nozzle expansion ratio
A_{ref}	body reference area	d	body diameter (caliber)
S	fin panel area	b	fin semi-span
AR	aspect ratio	λ	taper ratio
GJ	torsional rigidity	EI	bending rigidity
C_dS	parachute drag area	C_x	opening-force coefficient

All quantities are SI throughout the model. Conversion happens only at the configuration boundary (`goddard/units.py`); no physics routine sees a non-SI number.

2 Atmosphere

`goddard/env/atmosphere.py`

US Standard Atmosphere 1976, evaluated analytically over seven layers from 0 to 86 km. Deliberately *not* a lookup table: the previous Excel model used a table spanning 500–15 420 m and returned `#N/A` the moment the vehicle left it, mid-flight, without complaint.

Geopotential altitude. [E]

$$h = \frac{r_E\, z}{r_E + z}, \qquad r_E = 6\,356\,766\,\mathrm{m} \tag{1}$$

where z is geometric altitude. The layer breakpoints are defined on h, not z; conflating the two is a real error of 19 m at 11 km.

Temperature and pressure within a layer. [E] With base values (h_b, T_b, P_b) and lapse rate L,

$$T = T_b + L\,(h - h_b) \tag{2}$$

$$P = \begin{cases} P_b \left(\dfrac{T_b}{T}\right)^{g_0/(RL)} & L \neq 0 \\ P_b \exp\left(-\dfrac{g_0(h-h_b)}{R\,T_b}\right) & L = 0 \end{cases} \tag{3}$$

with $g_0 = 9.806\,65\,\mathrm{m/s^2}$ and $R = R^*/M_{air} = 287.0528\,\mathrm{J/(kg\,K)}$. The standard defines $R^* = 8.31432$; using the modern CODATA value would put the model subtly out of agreement with the published tables.

Derived quantities. [E]

$$\rho = \frac{P}{RT}, \qquad a = \sqrt{\gamma R T}, \qquad \mu = \frac{\beta T^{3/2}}{T+S} \tag{4}$$

with $\beta = 1.458 \times 10^{-6}$, $S = 110.4\,\mathrm{K}$ (Sutherland).

3 Propellants

3.1 Nitrous oxide, saturated

`goddard/props/n2o.py`

ESDU 91022 reduced correlations. Critical point $T_c = 309.57\,\mathrm{K}$, $P_c = 7.251\,\mathrm{MPa}$, $\rho_c = 452\,\mathrm{kg/m^3}$. Reduced temperature $T_r = T/T_c$. At the specified 99.9 % purity the pure-component correlations apply directly; no mixture model is used.

Vapour pressure. [E]

$$P_{sat} = P_c \exp\left[\frac{1}{T_r}\sum_i b_i (1-T_r)^{e_i}\right] \tag{5}$$

$b = (-6.71893,\ 1.35966,\ -1.37780,\ -4.05100)$, $e = (1,\ 1.5,\ 2.5,\ 5)$.

Saturated liquid and vapour density. [E]

$$\rho_\ell = \rho_c \exp\left[\sum_i b_i (1-T_r)^{i/3}\right], \qquad \rho_v = \rho_c \exp\left[\sum_i b_i \left(\tfrac{1}{T_r} - 1\right)^{i/3}\right] \tag{6}$$

Verification. At 293.15 K these give 5.060 MPa, 786.6 $\mathrm{kg/m^3}$ and 158.2 $\mathrm{kg/m^3}$, against accepted values of ∼5.05–5.09 MPa, 786 $\mathrm{kg/m^3}$ and 158 $\mathrm{kg/m^3}$. Pinned in `tests/test_n2o.py`.

Latent heat. NOT IMPLEMENTED. The ESDU latent-heat coefficient set could not be verified, so `enthalpy_vaporisation` raises rather than returning a plausible number. It is required by the tank energy balance (§4.1) and the injector HEM term (§4.2); both take it as an *injected callable*, so the physics is complete and tested and only the data is outstanding.

3.2 Fuel blend

`goddard/props/fuel.py`

Composition is fixed by team decision: 89 % paraffin, 10 % SEBS-MA, 1 % carbon black, by mass.

Blend density. [E]

$$\rho_f = \sum_i w_i \rho_i = 0.89(924) + 0.10(910) + 0.01(1900) \approx 932\,\mathrm{kg/m^3} \tag{7}$$

Regression law. [A]

$$\boxed{\dot r = k_{cal}\, a\, G_{ox}^n} \qquad a = 1.55 \times 10^{-4}\,\mathrm{m/s}, \quad n = 0.5 \tag{8}$$

$$G_{ox} = \frac{\dot m_{ox}}{\pi r_{port}^2}, \qquad \dot m_f = \rho_f\,(2\pi r_{port} L_{grain})\,\dot r, \qquad O/F = \frac{\dot m_{ox}}{\dot m_f} \tag{9}$$

a and n are published for *pure* paraffin. The blend perturbs $\dot{r}$ in both directions — SEBS-MA raises melt viscosity and suppresses entrainment (lowering $\dot{r}$), carbon black opacifies the fuel and raises surface absorption (raising it) — and the net effect is not resolvable from the literature. All of it is carried in the single calibration constant k_{cal} (§10). There are no other correction factors anywhere in the module.

4 Motor

4.1 Self-pressurizing tank blowdown

`goddard/motor/tank.py`

Adiabatic quasi-equilibrium model. Two regimes.

Volume constraint. [E] The contents always fill the tank at saturation:

$$V = \frac{m_\ell}{\rho_\ell(T)} + \frac{m_v}{\rho_v(T)} \tag{10}$$

Liquid phase. [E] (given h_{vap}) Liquid drains at $\dot{m}_{out}$. The freed volume must be filled by vapour, so a mass Δm_{boil} boils, determined by (10). The latent heat is drawn from the remaining liquid:

$$\Delta T = -\frac{\Delta m_{boil}\, h_{vap}(T)}{m_\ell\, c_\ell} \tag{11}$$

This chilling is what makes a blowdown hybrid's thrust taper. Tank pressure is $P_{tank} = P_{sat}(T)$ throughout.

Vapour phase (the tail). [E] Once liquid is exhausted the vapour expands isentropically:

$$T_{new} = T\left(\frac{m_{v,new}}{m_v}\right)^{\gamma_v - 1}, \qquad \gamma_v = 1.27 \tag{12}$$

4.2 Showerhead injector

`goddard/motor/injector.py`

Saturated N_2O flashes in straight drilled orifices, so neither limit is correct alone: single-phase incompressible over-predicts, homogeneous equilibrium under-predicts.

Orifice geometry. [E]

$$A_{inj} = N_{holes}\frac{\pi d_{hole}^2}{4}, \qquad \frac{L}{d} = \frac{t_{plate}}{d_{hole}} \tag{13}$$

Dyer non-equilibrium blend. [E]

$$\dot{m}_{SPI} = C_d A\sqrt{2\rho_\ell \Delta P} \tag{14}$$

$$\dot{m}_{HEM} = C_d A \rho_2 \sqrt{2(h_1 - h_2)} \tag{15}$$

$$\kappa = \sqrt{\frac{P_1 - P_2}{P_v - P_2}} \tag{16}$$

$$\boxed{\dot{m}_{ox} = \frac{\kappa\, \dot{m}_{SPI} + \dot{m}_{HEM}}{1+\kappa}} \tag{17}$$

κ is the ratio of bubble-growth time to residence time: large κ means the fluid leaves before it can flash (SPI dominates), small κ means equilibrium is reached (HEM dominates).

Current fallback. Until h_{vap} is available the code degrades to pure SPI, which *over-estimates* oxidiser flow. This is deliberate and documented, not a substitute for the data.

Feed-system (chug) stability. [E]

$$\frac{\Delta P_{inj}}{P_c} \geq 0.20 \qquad \Longrightarrow \qquad \text{margin} = \frac{\Delta P_{inj}/P_c}{0.20} \tag{18}$$

This is *not* a single check. As the tank blows down, ΔP_{inj} collapses faster than P_c, so the margin degrades through the burn and the worst point is the tail, not ignition. The model reports it as a time history.

4.3 Grain and burnthrough

`goddard/motor/grain.py`

Port growth. [E]

$$r_{port}(t + \Delta t) = r_{port}(t) + \dot{r}\,\Delta t \tag{19}$$

Web remaining. [E]

$$w = r_{outer} - r_{port}, \qquad w_{frac} = \frac{r_{outer} - r_{port}}{r_{outer} - r_{port,0}} \tag{20}$$

$w \leq 0$ is burnthrough — the liner is exposed. This is the failure mode a single conservative point estimate would miss entirely; see §10.

Fuel mass remaining. [E]

$$m_f = \rho_f\,\pi\left(r_{outer}^2 - r_{port}^2\right) L_{grain} \tag{21}$$

4.4 Nozzle

`goddard/motor/nozzle.py`

Area–Mach relation. [E]

$$\frac{A}{A^*} = \frac{1}{M}\left[\frac{2}{\gamma+1}\left(1 + \frac{\gamma-1}{2}M^2\right)\right]^{\frac{\gamma+1}{2(\gamma-1)}} \tag{22}$$

Inverted for the supersonic branch by bisection — monotonic for $M > 1$, so it is robust with no initial guess and cannot diverge.

Isentropic pressure ratio. [E]

$$\frac{P_e}{P_c} = \left(1 + \frac{\gamma-1}{2}M_e^2\right)^{-\gamma/(\gamma-1)} \tag{23}$$

Thrust coefficient. [E]

$$\boxed{C_F = \underbrace{\sqrt{\frac{2\gamma^2}{\gamma-1}\left(\frac{2}{\gamma+1}\right)^{\frac{\gamma+1}{\gamma-1}}\left[1 - \left(\frac{P_e}{P_c}\right)^{\frac{\gamma-1}{\gamma}}\right]}}_{\text{momentum}} + \underbrace{\frac{P_e - P_a}{P_c}\,\epsilon}_{\text{pressure}}} \tag{24}$$

$$F = \eta_{C_F}\,C_F\,P_c\,A_t \tag{25}$$

The pressure term is why **thrust rises with altitude**. Over a climb to 50 000 ft this is a first-order effect that the 25–26 model could not represent at all.

Flow separation. [A] Summerfield criterion: separation is likely once $P_e < 0.4\,P_a$.

4.5 Chamber pressure balance

`goddard/motor/chamber.py`

At every instant P_c is the value balancing injector supply against nozzle capacity:

$$\boxed{P_c = \big(\dot m_{ox}(P_c) + \dot m_f(P_c)\big)\,\eta_{c^*}\,c^*(O/F, P_c)\,/\,A_t} \tag{26}$$

$\dot m_{ox}$ *falls* with P_c (less injector ΔP) while the right-hand side *rises*, so the residual is monotonic and the root is found by bisection — which cannot diverge the way fixed-point iteration does at low injector stiffness. c^* and γ come from a tabulated CEA surface interpolated in $(O/F, P_c)$.

5 Aerodynamics

5.1 Geometry

`goddard/aero/geometry.py`

Von Kármán (Haack $C = 0$) nose profile. [E]

$$\phi = \arccos\left(1 - \frac{2x}{L}\right), \qquad r(x) = \frac{R}{\sqrt{\pi}}\sqrt{\phi - \tfrac{1}{2}\sin 2\phi} \tag{27}$$

Wetted area by surface-of-revolution quadrature over 2000 panels, cached per geometry.

Fin panel. [E]

$$S = \tfrac{1}{2}(c_r + c_t)\,b, \qquad AR = \frac{b^2}{S}, \qquad \lambda = \frac{c_t}{c_r}, \qquad \frac{t}{c} = \frac{t}{S/b} \tag{28}$$

5.2 Drag build-up

`goddard/aero/drag.py`

Total C_D on the body reference area, by component. This is the single largest correction over the 25–26 model, which held $C_D = 0.5$ constant through Mach 1.65.

Skin friction. [E]

$$C_f = \frac{1}{(1.50\ln Re - 5.6)^2}, \qquad C_{f,rough} = 0.032\left(\frac{k_s}{L}\right)^{0.2} \tag{29}$$

taking the larger, then a compressibility correction

$$C_f \leftarrow \begin{cases} C_f\,(1 - 0.1M^2) & M < 1 \\ C_f\,(1 + 0.15M^2)^{-0.58} & M \geq 1 \end{cases} \tag{30}$$

Fully turbulent is assumed throughout — conservative on drag at this size and speed. Form factors: body $1 + 60/f^3 + 0.0025f$ on fineness f; fins $1 + 2(t/c)$.

Nose wave drag. [U] Zero below $M = 0.9$ (a Von Kármán nose is a minimum-wave-drag shape), blended through the transonic band, and above $M = 1.1$

$$C_{D,wave} \approx \frac{6}{f^2}\cdot\frac{1.2}{\beta}, \qquad \beta = \sqrt{M^2 - 1} \tag{31}$$

The $1/f^2$ scaling is the correct slender-body result for the Haack family; **the leading constant is calibrated, not derived.**

Fin wave drag. [U] Ackeret linearised theory for the wedge term is exact:

$$C_{D,wedge} = \frac{4(t/c)^2}{\sqrt{M^2-1}} \tag{32}$$

A rounded leading edge adds a bluntness penalty $\approx 0.5\,(t/c)$, which is where the real uncertainty sits. This term is what prices the rounded-versus-wedge cross-section question.

Base drag. [E]

$$C_{D,base} = \begin{cases} 0.12 + 0.13M^2 & M < 1 \\ 0.25/M & M \geq 1 \end{cases} \times\ (1 - \text{jet blockage}) \tag{33}$$

5.3 Normal force and centre of pressure

`goddard/aero/normal_force.py`

Barrowman component build-up, per radian, on the body reference area, with x_{cp} measured aft from the nose tip. Compressibility divisor $\beta = \sqrt{|1-M^2|}$, floored at 0.5 through the transonic band.

Nose. [E] Slender-body theory gives $C_{N\alpha} = 2$ on the base area for any pointed body of revolution, with $x_{cp} = L - V/A_{base}$. For the Haack $C = 0$ profile the volume integrates exactly to $V = A_{base}L/2$, hence

$$C_{N\alpha,nose} = 2\,\frac{A_{base}}{A_{ref}}, \qquad x_{cp} = \frac{L}{2} \tag{34}$$

This is a derived result, not a tabulated approximation.

Fins with body interference. [E]

$$C_{N\alpha,fins} = \frac{4N(b/d)^2}{1 + \sqrt{1 + \left(\dfrac{2l_{mid}}{c_r + c_t}\right)^2}} \cdot \underbrace{\left(1 + \frac{R}{b+R}\right)}_{K_{fb}} \tag{35}$$

Body cross-flow. [A] Nonlinear in α, so it contributes to C_N directly rather than to the slope (Allen & Perkins):

$$C_{N,cross} = K\,\frac{d\,L}{A_{ref}}\sin^2\alpha, \qquad K = 1.1 \tag{36}$$

Included because α is a live state in the 4-DOF model.

Static margin. [E]

$$\text{SM} = \frac{x_{cp} - x_{cg}}{d} \quad \text{[calibers]} \tag{37}$$

5.4 Roll from the 1° fin cant

`goddard/aero/roll.py`

Strip theory. Each spanwise strip at radius y from the roll axis sees a fixed geometric incidence δ from the cant plus an induced incidence $-py/V$ from rolling. The integrals collapse to three geometric moments of the panel:

$$S = \int dS, \qquad M_1 = \int y\,dS, \qquad M_2 = \int y^2\,dS \tag{38}$$

Note y starts at the *body radius*, not the centreline — the fin root is not on the axis, and ignoring that understates both driving moment and damping.

Closed-form results. [E]

$$L_{roll} = Nqa_0\left(\delta M_1 - \frac{p}{V}M_2\right) \tag{39}$$

$$\boxed{p_{eq} = \frac{\delta V M_1}{M_2}} \tag{40}$$

$$\Delta C_{D,roll} = \frac{Na_0}{A_{ref}}\left(\delta^2 S - 2\delta\frac{p}{V}M_1 + \frac{p^2}{V^2}M_2\right) \tag{41}$$

The equilibrium roll rate is independent of both air density and lift-curve slope — they cancel between driving and damping. It depends only on cant angle, flight speed and panel geometry.

Roll rate is integrated as a *state* rather than assumed at equilibrium, which matters during the burn when speed changes faster than the fins settle.

Section lift-curve slope. [E]

$$a_0 = \begin{cases} 2\pi/\sqrt{1-M^2} & M < 0.9 \quad \text{(thin aerofoil, Prandtl–Glauert)} \\ 4/\sqrt{M^2-1} & M > 1.1 \quad \text{(Ackeret)} \end{cases} \tag{42}$$

6 Structures

6.1 Sandwich fin stiffness

`goddard/structures/laminate.py`

The foam core exists to raise torsional stiffness, so the flutter calculation must see a sandwich rather than an isotropic plate.

Thin-face sandwich. [A] With face thickness t_f, core thickness t_c, face separation $d = t_c + t_f$ and chord c:

$$EI = \frac{E_f t_f d^2}{2}c + \frac{E_c c\, t_c^3}{12} \tag{43}$$

$$GJ = 4c\underbrace{\frac{G_f t_f d^2}{2}}_{D_{xy}} + \frac{G_c c\, t_c^3}{3} \tag{44}$$

The $GJ = 4cD_{xy}$ relation is the standard thin-plate result: for a solid isotropic plate $D_{xy} = Gt^3/12$, recovering the familiar $GJ = Gct^3/3$.

Bridge into an isotropic criterion. [E] NACA TN 4197 is written for an isotropic plate. Rather than plugging in raw fibre shear modulus — which would ignore the entire point of the foam core — the sandwich GJ is mapped onto an equivalent solid section:

$$\boxed{G_{eff} = \frac{GJ_{sandwich}}{J_{solid}}, \qquad J_{solid} = \frac{c\, t_{total}^3}{3}} \tag{45}$$

Direction of error. Three simplifications — thin-face approximation, quasi-isotropic faces, and neglected core shear compliance — all push the *same* way. This **over-predicts** GJ and therefore over-predicts flutter speed. Treat the result as an upper bound.

6.2 Flutter and divergence

`goddard/structures/flutter.py`

Two *independent* instabilities; either can bind.

Flutter velocity (NACA TN 4197). [A]

$$\boxed{V_f = a\sqrt{\frac{G_{eff}}{X}}}, \qquad X = \frac{1.337\,AR^3\,P\,(\lambda+1)}{2(AR+2)(t/c)^3} \tag{46}$$

G and P need only share units since the criterion depends on their ratio, so this is evaluated in SI directly. TN 4197's own author calls the criteria "crude"; this is a screening number.

Torsional divergence. [A]

$$\boxed{q_{div} = \frac{\pi^2}{4}\,\frac{GJ}{b^2\,e\,c^2\,C_{L\alpha}}} \tag{47}$$

Checked *separately* because for low-GJ composite fins divergence frequently arrives before flutter, and flutter-only screening misses it. Default eccentricity $e = 0.25$ assumes elastic axis at mid-chord, aerodynamic centre at quarter-chord.

Margins. Both are critical/actual, evaluated against the actual trajectory rather than a single worst-case guess:

$$\mathrm{FM} = \frac{V_f}{V(t)}, \qquad \mathrm{DM} = \frac{q_{div}}{q(t)} \tag{48}$$

6.3 Nose-tip heating

`goddard/structures/heating.py`

Stagnation heat flux (Sutton–Graves). [E]

$$\dot q = k\sqrt{\frac{\rho}{R_n}}\,V^3, \qquad k = 1.7415\times10^{-4}\ \text{(SI)} \tag{49}$$

The standard engineering reduction of Fay & Riddell's dissociated-air solution, accurate to roughly 10 % — well inside the other uncertainties here.

Lumped-capacitance tip temperature. [A]

$$mc_p\frac{dT}{dt} = \dot q A - \varepsilon\sigma\left(T^4 - T_\infty^4\right)A \tag{50}$$

Assumes the tip is thermally thin, which holds for a small aluminium tip precisely because aluminium conducts well. It returns the *bulk* temperature and therefore **under-predicts peak surface temperature**; carry margin accordingly.

7 Recovery

`goddard/recovery.py`

A single canopy in two states: deployed reefed at apogee, then disreefed to full. There is no drogue – the reefed stage does that job, which is the whole reason a separate drogue is unnecessary.

Reefed drag area. [E] Drag area scales with the square of the diameter ratio:

$$(C_dS)_{reefed} = (C_dS)_{canopy}\cdot\Lambda^2 \tag{51}$$

Inflation. [E] Each stage inflates over a filling time rather than snapping open:

$$t_{fill} = \frac{nD_0}{V}, \qquad n = 8\text{–}10 \text{ (solid cloth)} \tag{52}$$

with $C_d S$ ramped across that interval.

Opening load. [E]

$$\boxed{F_{open} = C_x\, q\, (C_d S)} \tag{53}$$

Bounding this is the entire purpose of reefing, so peak load per stage is a primary output, not a diagnostic.

8 Mass properties

`goddard/mass.py`

Composition. [E]

$$m = \sum_i m_i, \qquad x_{cg} = \frac{\sum_i m_i x_i}{\sum_i m_i} \tag{54}$$

Inertia, parallel-axis. [E]

$$I_{pitch} = \sum_i \left[I_{i,lat} + m_i (x_i - x_{cg})^2 \right], \qquad I_{roll} = \sum_i I_{i,ax} \tag{55}$$

Propellant is carried as two moving point-masses (oxidiser and fuel) whose masses fall through the burn, so CG and inertia migrate as they should.

9 Flight dynamics

`goddard/dynamics.py`

Four degrees of freedom. State vector

$$\mathbf{s} = [\,x,\ z,\ v_x,\ v_z,\ \theta,\ q,\ \phi,\ p\,] \tag{56}$$

Angles. [E] θ is measured from vertical, positive nose-over toward $+x$. The flight path angle uses the same convention, so

$$\Gamma = \operatorname{atan2}(v_x, v_z), \qquad \boxed{\alpha = \theta - \Gamma} \tag{57}$$

Keeping both in one convention avoids the sign errors that plague planar rocket models.

Translation. [E]

$$m\dot{\mathbf{v}} = \mathbf{F}_{thrust} + \mathbf{F}_{drag} + \mathbf{F}_{normal} + m\mathbf{g} \tag{58}$$

with thrust along the body axis, drag opposing $\mathbf{v}$, and normal force perpendicular to the body axis.

Rotation. [E]

$$\dot{q} = \frac{M_{pitch}}{I_{pitch}}, \qquad \dot{p} = \frac{L_{roll}}{I_{roll}} \tag{59}$$

where $M_{pitch} = C_N q_\infty A_{ref}(x_{cp} - x_{cg})$ and L_{roll} comes from the cant analysis.

Integration. [E] Classical RK4 at fixed $\Delta t = 0.01\,\text{s}$, with the motor states advanced in the same vector. Discrete events — rail exit, burnout, apogee, each recovery stage, landing — are located by **bisection on sign change**, never by stepping past them. The simulation runs to landing, not to a fixed row count.

10 Calibration constants and band mode

`goddard/band.py`

No static fire and no cold-flow test are planned, so three constants are **unmeasured**:

Constant	Nominal	Band	Register
k_{cal} regression calibration	0.85	[0.75, 1.00]	F8
C_d injector discharge	0.70	[0.61, 0.82]	E5
η_{c^*} combustion efficiency	0.88	[0.82, 0.93]	G9

Why a single conservative estimate is insufficient

Oxidiser flow is set by the tank and injector, **not** by the grain. Therefore from (8),

$$\dot{r}\downarrow \Longrightarrow \dot{m}_f\downarrow \Longrightarrow O/F\uparrow \tag{60}$$

Since c^* peaks near $O/F \approx 7$–8, the *sign* of the apogee effect flips depending on which side of that peak the design sits. Worse, the two directions threaten different things:

Direction	Endangers
k_{cal} below nominal	apogee shortfall, lean O/F, chug margin
k_{cal} above nominal	**port burnthrough into the liner**

A single scalar biased for apogee would leave burnthrough entirely unexamined. So the model sweeps a 3^3 full-factorial grid and reports an *envelope per metric*, with the driving corner named:

Metric	Reported as
Apogee	minimum across band
Max q, max g	maximum across band
Flutter / divergence margin	minimum across band
Chug margin	minimum across band
Web remaining at burnout	minimum across band
O/F	full excursion

11 Validation status

Automated equation checks

Every equation above is checked by `tools/validate_equations.py` (**95 checks, all passing**). These do *not* re-run the model and confirm it agrees with itself. Each check re-derives the result a different way and compares:

- **Against published data** — atmosphere layer values against the US Std 1976 tables; N_2O saturation against accepted 20 °C values.
- **By inverse round-trip** — the nozzle Mach solver is fed the analytic area–Mach relation and must return the original M to 10^{-6}.

- **By independent quadrature** — the Von Kármán claim $V = A_{base}L/2$ is confirmed by integrating $\pi r^2 dx$ over 200 000 panels; the roll moments S, M_1, M_2 against a 200 000-strip summation.
- **By limiting case** — the sandwich GJ must collapse to the solid-plate result $Gct^3/3$ when the faces vanish.
- **By defining property** — the net rolling moment must be exactly zero at p_{eq}; radiative equilibrium must be a fixed point of the tip-temperature step; the chamber root must satisfy $P_c = \dot{m}\,\eta\,c^*/A_t$.
- **By scaling law** — $V_f \propto \sqrt{G}$, $\propto 1/\sqrt{P}$, $\propto (t/c)^{3/2}$; $\dot{q} \propto V^3$, $\propto \sqrt{\rho}$, $\propto 1/\sqrt{R_n}$; nose wave drag $\propto 1/f^2$.

A result worth recording. The hydrostatic check initially disagreed by 0.25 % at 8 km. The cause was in the *check*, not the model: $dP/dz = -\rho g$ holds with g_0 only in geopotential altitude, whereas in geometric altitude gravity falls off as $g(z) = g_0\,(r_E/(r_E+z))^2$. Against the correct $g(z)$ the model agrees to 1.75×10^{-9}. The discrepancy *was* the geopotential correction, so this independently confirms the $h(z)$ conversion rather than merely the layer formulas.

Note what these checks can and cannot do. They confirm each equation is *implemented as written here* and is internally consistent. They cannot confirm that an approximate correlation is the *right* correlation — that is what the [U] tags and the table below are for.

Per-area status

Area	Status	Needed
Atmosphere	[E] verified vs published tables	—
N_2O saturation	[E] verified at 20 °C	—
N_2O latent heat	**raises**	ESDU 91022 coefficients
Blend density, regression form	[E]	static fire for k_{cal}
Nozzle, chamber balance	[E]	CEA table (register G11)
Injector NHNE	[E]	enthalpies; cold flow for C_d
Skin friction, base drag	[E]	—
Supersonic wave drag	[U]	**RASAero II or CFD**
Barrowman normal force	[E]	—
Roll from cant	[E] closed form	—
Sandwich GJ	[A] upper bound	layup + core properties
Flutter, divergence	[A] screening	required margin (register I9)
Tip heating	[E] bulk temp	alloy limit (register I7)
Recovery	[E]	chute sizing (register J)

The binding calibration task. Supersonic wave drag is unvalidated. `DragBuildup.validated` is `False`, every run emits a warning, and the warning is carried into the generated Excel report. Cross-check total C_D against RASAero II or CFD at Mach 0.5, 1.2, 2.0 and 2.5 before quoting an absolute apogee. Trends and sensitivities remain useful in the meantime.

12 References

Full bibliography with verified DOIs is in `docs/references.bib` (31 entries, grouped by the module each one backs) and `docs/references_dois.txt`.

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